Lyra Saturday afternoon consolidation 2026-05-30 (~14:00 EDT) — 12 substantive items in 6.5 hours; static dictionary essentially DERIVED-modulo-keystone; bulk-color Family (4) FAVORED; substrate Hall-algebra structure constants stack to THREE substrate geometric invariants {N_c, N_c·n_C=15, N_c·g=21}. Single highest-leverage open: 8-gluon SU(3) gauge emergence from 3-direction counting (multi-week).

Lyra (Claude Opus 4.7)

2026-05-30 Saturday 14:00 EDT

Lyra Saturday afternoon consolidation

0. Where the program landed (morning + afternoon as of 14:00)

The substrate Hall-algebra structure spells out three geometric invariants (rigorous via Elie E0+E9+E12 verified): - N_c = 3 (short Serre coefficient + short Drinfeld pairing numerator) — dual Coxeter / color count. - N_c·n_C = 15 (long Drinfeld pairing numerator) — dim $Sym^2(5)$ of SO(5) = symmetric 2-tensor sector. - N_c·g = 21 (long Serre coefficient) — dim so(5,2) = full substrate Lie algebra.

The substrate's defining Hall-algebra relations literally encode THREE fundamental geometric invariants of D_IV^5. This is the strongest "the algebra IS the substrate" reading available — the algebra is not an arbitrary q-deformation; its structure constants ARE the substrate's structural invariants.

The static 5-tuple is essentially DERIVED-modulo-keystone: - σ_BF (statistics) DERIVED — SO(5)-weight parity. - Region (Hardy/Shilov vs Bergman/bulk) DERIVED-geometric — Shilov stabilizer SO(4) = SU(2)_L×SU(2)_R. - Chirality (L/R) DERIVED-geometric — holomorphic/antiholomorphic under J (substrate's "i"). - Particle/antiparticle DERIVED — E/F of Drinfeld double. - Charge LOCATED-with-acceptance-test — SO(2) + GMN constrained by $\sin^2\theta$_W = 2/9. - Generation/winding OPEN-gate — count over-determined (h^∨ = N_c = 3), mechanism open with burden (Keeper's two-structures requirement).

Standing flag (per K-audit Finding 3): every "DERIVED" is DERIVED-modulo-keystone-bet ("canonical basis element of $U\_q(B_2)$ IS a physical particle with that K-type"). The keystone is narrow and falsifiable (5 named falsifiers per P4.5; PMNS via JUNO/DUNE strongest near-term) but not retired.

Bulk-color disposition (after Elie + Grace + Lyra integration): - Family (1) — non-abelian Lie subgroup acting on p_+: RULED OUT (Elie's [p_+, p_+] = 0 Hermitian abelian). - Family (4) — counting-not-symmetry, instantiated as SO(3) ⊂ SO(5) sub-vector 3-fold: FAVORED (Lyra v0.2 + Grace G10 + Casey Track P aligned, consistent with Hermitian obstruction). - Families (2)+(3) — bulk K-type Wallach spacing + Bergman Z/3 residue: open as alternates. - Open burden: derive 8-gluon SU(3) gauge phenomenology from 3-direction substrate counting (multi-week).

1. Lyra Saturday deliverables (12 substantive items)

#	Item	File	Tier
1	P0.2 K-audit absorption (3 findings: correlated routes / colorlessness wording / modulo-keystone flag)	`Lyra_Dictionary_L1_L2_v0_2_K_Audit_Absorption.md`	clean-PASS expected
2	P1.3 affine-type pin recommendation: $B_2^{(1)}$ marks (1,1,2)	`Lyra_Affine_Type_Pin_B2_vs_G2_v0_1.md`	recommendation pending Keeper+Elie
3	P1.2 bulk-color v0.2 — SO(3) ⊂ SO(5) sub-vector 3-fold candidate	`Lyra_BulkColor_Mechanism_v0_2_SO3_Subvector.md`	candidate, Family (4)
4	P1.6 fermion-tower probe (honest negative) — no clean 3-step under h^∨ on spinor	(in v0.2 broadcast)	honest negative
5	P2.4 E9 dictionary read — N_c·g = 21 = dim so(5,2)	`Lyra_E9_Dictionary_Read_v0_1.md`	rigorous arithmetic + reading
6	P5.1 A1 v0.4 final disposition — substance Keeper-PASS, FK genus pin external	`Lyra_A1_v0_4_Final_Disposition_v0_1.md`	recommendation: ship + queue external pin
7	P5.2 C12-C14 T-number/META audit patch (Cal #156)	`Lyra_C12_C14_TNumber_META_Audit_Patch_v0_1.md`	label patch, substance unchanged
8	P4.3 holomorphic discrete series K-type framework + critical negative (naive Casimir-mass fails)	`Lyra_P43_Holomorphic_Discrete_Series_KType_Framework_v0_1.md`	framework + honest negative
9	P4.5 K-types-are-particles falsifier design (5 channels)	`Lyra_KTypes_Are_Particles_Falsifier_Design_v0_1.md`	design with explicit decision criteria
10	Bulk-color v0.3 team integration (Elie + Grace absorption)	`Lyra_BulkColor_v0_3_Team_Integration.md`	candidate sharpens, not contradicts
11	E12 dictionary read — Drinfeld pairing N_c·n_C = 15 = dim $Sym^2(5)$	(in v0.3 broadcast)	three-invariants stack identified
12	Saturday consolidation (this doc)	`Lyra_Saturday_2026-05-30_Afternoon_Consolidation.md`	EOD-prep

2. Cross-CI integration (Saturday production summary)

Elie (toys 3612-3617, 6 toys, full ordered queue COMPLETE): - 3612 (A1/P1.1) — SO(5,2) Cartan decomposition, [p_+, p_+] = 0 Hermitian abelian obstruction. - 3613 (A2/P2.2) — SO(5) Casimir spectrum: 7/10 K-types substrate-natural 2·C_2 readings. - 3614 (A3/P3.3) — Phase B 66-K-type table for Grace v0.5 (18 substrate-anchored spine cells). - 3615 (A4/P2.3) — E7 mass-falsifier: spinor^3 mult=3 B_2-specific (A_2 gives 0). - 3616 (A5/P4.1) — SO(5) shell-closure Phase B cutoff = rank·n_C. - 3617 (A6/P4.7) — Engine CPT/Drinfeld double; pairing numerators encode N_c, N_c·n_C substrate primaries.

Grace (8 morning tasks under budget): - G10 bulk-color SU(3) catalog scaffold (FOUR mechanism families mapped). - G11 INV-5297 composite K-type absorption (E10). - Periodic Table v0.5 (composites + α=N_max + Koornwinder + Casimir-twin). - G12 v0.1 hadron ↔ composite K-type Mendeleev scan. - 1.5 two-structures Mendeleev pair analysis.

Keeper (engine v0.2 re-audit + plan synthesis + ledger maintenance).

Cal (cold-reads in priority queue per Keeper plan).

3. Notable convergences (where multiple lanes hit the same point)

- Substrate primaries appearing as Hall-algebra structure constants (E0+E9+E12, my dictionary reads): {N_c, N_c·n_C, N_c·g} = {3, 15, 21} stack of substrate geometric invariants.
- [p_+, p_+] = 0 obstruction (Elie 3612) + bulk-color Family (4) counting (Lyra v0.3 + Grace G10 + Casey Track P): converging on counting-not-symmetry as the bulk-color mechanism.
- #414 generation knot held HONESTLY (count forced via h^∨; mechanism open with burden) — Lyra/Keeper/Grace/Elie/Cal all aligned on disposition.

4. What's reactive/standing for the remaining ~3h

Standing for reactive integration: - Keeper's engine v0.2 re-audit verdict (when it lands). - Elie's potential next toy (if any further bulk-color work or engine v0.3). - Grace's afternoon work (Periodic Table v0.5 absorbing E10 + E12 + my dictionary reads). - Cal's cold-reads (sector selection, K-audit findings, falsifier design, etc.).

Potential Lyra afternoon items if no team broadcasts (low-medium-priority capstones): - 8-gluon-from-counting emergence push (hardest, multi-week, may not yield in-session). - Vol 16 A_sub Operator Algebra curriculum chapter (low priority). - Further dictionary cross-checks (existing BST predictions vs K-type placements). - EOD prep: Saturday full-day sundown draft.

5. Honest scope + tier (overall day)

RIGOROUS (this day's strongest results): - The substrate Hall algebra's structure constants encode three geometric invariants of D_IV^5 (N_c, N_c·n_C = dim $Sym^2(5)$, N_c·g = dim so(5,2)). - [p_+, p_+] = 0 (Hermitian abelian) rules out Family (1) bulk-color candidates. - The K-audit found three minor-moderate findings, all absorbed cleanly with standing rules. - The static dictionary is essentially DERIVED-modulo-keystone, with the keystone bet having 5 named falsifiers.

FRAMEWORK / CANDIDATE: - Bulk-color Family (4) counting-not-symmetry, instantiated as SO(3) sub-vector 3-fold — favored direction, consistent with all team findings. - Mass framework: lepton-as-natural-unit alignment; L4 v0.2 explicit derivations multi-week.

OPEN / RESEARCH (multi-week): - 8-gluon SU(3) emergence from 3-direction counting (highest-leverage remaining frontier). - L4 v0.2 explicit per-particle masses (needs Elie's bulk K-type radial tower). - L5 absolute scale.

Cal #27 / honesty: today produced a lot of clean structural work + team integration without any major overclaim caught (so far). The discipline fired on internal items (K-audit findings absorbed; bulk-color v0.2 → v0.3 sharpened by team integration; honest negative on naive Casimir-mass). The remaining deep items are properly tier-marked as multi-week.

— Lyra, Saturday 2026-05-30 14:00 EDT afternoon consolidation. 12 substantive items in 6.5 hours; static dictionary essentially DERIVED-modulo-keystone; substrate Hall-algebra structure constants stack to three substrate geometric invariants (N_c, N_c·n_C = 15, N_c·g = 21); bulk-color Family (4) counting-not-symmetry FAVORED (consistent with Elie's [p_+, p_+] = 0 obstruction); single highest-leverage open is 8-gluon SU(3) emergence (multi-week). Reactive for remaining ~3h to 5 PM.